

Hooks System

Introduced: v2.0.x (Late 2025) | Frontmatter support: v2.1.0 | Latest: v2.1.76

What It Is

Hooks are event-driven automations that fire at specific points in Claude Code’s lifecycle. When Claude is about to run a bash command, edit a file, or finish a response — hooks can intercept, validate, transform, or block the action. They’re configured in `settings.json` and can be shell commands, LLM prompts, or full subagents.

Why It Matters

Hooks turn Claude Code from interactive to automated:

- Auto-format files after every edit (Prettier, Black, gofmt)
- Block edits to protected files (`.env`, `package-lock.json`)
- Send desktop notifications when Claude needs attention
- Enforce quality gates before marking tasks complete
- Audit configuration changes for security
- Re-inject context after compaction

Quick Start (60 Seconds)

Add to `~/.claude/settings.json`:

```
{
  "hooks": {
    "Notification": [
      {
        "matcher": "",
        "hooks": [
          {
            "type": "command",
            "command": "osascript -e 'display notification \"Claude needs attention\" with tit"
          }
        ]
      }
    ]
  }
}
```

Now you get macOS notifications whenever Claude is waiting for input.

All Hook Events (22 Events)

Session Lifecycle

Event	Fires when	Can block?	Matcher values
SessionStart	Session begins/resumes	Yes	startup, resume, clear, compact
SessionEnd	Session terminating	Yes	clear, logout, prompt_input_exit, other

Event	Fires when	Can block?	Matcher values
PreCompact	Before context compaction	Yes	manual, auto
PostCompact	After context compaction completes	No	manual, auto

User Input

Event	Fires when	Can block?	Matcher
UserPromptSubmit	Before prompt processing	Yes	—

Tool Lifecycle

Event	Fires when	Can block?	Matcher values
PreToolUse	Before tool executes	Yes	Tool name: Bash, Edit, Write, mcp__server__tool
PostToolUse	After tool succeeds	No	Tool name
PostToolUseFailure	After tool fails	No	Tool name
PermissionRequest	Permission dialog	Yes	Tool name

Agent Lifecycle

Event	Fires when	Can block?	Matcher values
SubagentStart	Subagent spawned	No	Agent type: Explore, Plan, code-reviewer
SubagentStop	Subagent finished	No	Agent type
TeammateIdle	Teammate about to idle	Yes	—
TaskCompleted	Task marked complete	Yes	—

MCP Lifecycle

Event	Fires when	Can block?	Matcher values
Elicitation	MCP server requests structured input	Yes	MCP server name
ElicitationResponse	Before response sent back to MCP	Yes	MCP server name

System Events

Event	Fires when	Can block?	Matcher values
Stop	Claude finishes responding	Yes	—
Notification	Notification queued	No	permission_prompt, idle_prompt, auth_success

Event	Fires when	Can block?	Matcher values
ConfigChange	Config file modified	Yes	user_settings, project_settings, local_settings
WorktreeCreate	Worktree being created	Yes	—
WorktreeRemove	Worktree being removed	No	—

Hook Types

Type	What it is	Tool access?	Timeout	Best for
command	Shell command	No (stdin/stdout)	10 min	Formatting, validation, logging
prompt	Single LLM turn (Haiku)	No	10 sec	Policy decisions
agent	Multi-turn subagent	Yes	60 sec	Complex verification

Configuration

In settings.json

```
{
  "hooks": {
    "PreToolUse": [
      {
        "matcher": "Bash",
        "hooks": [
          {
            "type": "command",
            "command": "echo 'Bash command about to run'"
          }
        ]
      }
    ]
  }
}
```

Config Scope

File	Scope
~/.claude/settings.json	All projects
.claude/settings.json	This project (team-shared)
.claude/settings.local.json	Personal overrides
Skill/agent frontmatter	Scoped to skill lifecycle

Matcher Patterns (Regex)

```
"matcher": "Bash"           // Exact: Bash tool only
"matcher": "Edit|Write"     // Regex OR: Edit or Write
"matcher": "mcp__.*"        // All MCP tools
"matcher": "mcp__github__.*" // GitHub MCP tools only
"matcher": ""               // All (empty = match everything)
```

Exit Code Protocol

Exit Code	Effect
0	Allow. stdout added to Claude's context
2	Block. stderr shown as reason
Other	Allow. stderr logged in verbose mode only

Hooks receive JSON on stdin with event details and tool input.

Real-World Workflows

Workflow 1: Auto-Format After Every Edit

```
{
  "hooks": {
    "PostToolUse": [
      {
        "matcher": "Edit|Write",
        "hooks": [
          {
            "type": "command",
            "command": "jq -r '.tool_input.file_path' | xargs npx prettier --write 2>/dev/null"
          }
        ]
      }
    ]
  }
}
```

Workflow 2: Block Protected Files

```
{
  "hooks": {
    "PreToolUse": [
      {
        "matcher": "Edit|Write",
        "hooks": [
          {
            "type": "command",
            "command": "FILE=$(jq -r '.tool_input.file_path'); case \"$FILE\" in *.env*|*crede"
          }
        ]
      }
    ]
  }
}
```

```

    ]
  }
}

```

Workflow 3: Re-inject Context on Compaction

```

{
  "hooks": {
    "SessionStart": [
      {
        "matcher": "compact",
        "hooks": [
          {
            "type": "command",
            "command": "echo 'REMINDER: Use Bun not npm. Run bun test before commits. Current"
          }
        ]
      }
    ]
  }
}

```

Workflow 4: Quality Gate with Agent Hook

```

{
  "hooks": {
    "Stop": [
      {
        "hooks": [
          {
            "type": "prompt",
            "prompt": "Are all requested tasks complete? Return {\"ok\": true} if yes, {\"ok\"}"
          }
        ]
      }
    ]
  }
}

```

Workflow 5: Re-inject Context After Compaction (PostCompact)

```

{
  "hooks": {
    "PostCompact": [
      {
        "hooks": [
          {
            "type": "command",
            "command": "echo 'CONTEXT RESTORED: Key project details – use Bun, run tests before"
          }
        ]
      }
    ]
  }
}

```

```

    ]
  }
}

```

Workflow 6: Intercept MCP Elicitation

```

{
  "hooks": {
    "Elicitation": [
      {
        "matcher": "sentry",
        "hooks": [
          {
            "type": "command",
            "command": "echo 'MCP server requesting input - logging for audit' >> ~/mcp-elicitation.log"
          }
        ]
      }
    ]
  }
}

```

Workflow 7: Audit Config Changes

```

{
  "hooks": {
    "ConfigChange": [
      {
        "matcher": "",
        "hooks": [
          {
            "type": "command",
            "command": "echo \"$(date): config changed\" >> ~/claude-config-audit.log"
          }
        ]
      }
    ]
  }
}

```

How It Interacts With Other Features

Feature	Interaction
Skills	Skills can define hooks in frontmatter (active only during skill)
Worktrees	WorktreeCreate/WorktreeRemove hooks automate setup/cleanup
Agent Teams	TeammateIdle and TaskCompleted enforce quality gates
MCP	PreToolUse can match mcp__server__tool patterns
Subagents	SubagentStop includes last_assistant_message field
Permissions	PreToolUse hooks are separate from permission rules (both can deny)

Feature	Interaction
Plugins	Plugins can define hooks in <code>hooks/hooks.json</code>

Gotchas & Tips

1. **Shell profile breaks JSON** — if `.zshrc` has echo statements, they prepend to hook output. Guard with `if [[ $- == *i* ]]`.
2. **Exit 2 blocks AND shows reason** — `stderr` becomes the deny message. Exit 0 to add context without blocking.
3. **Stop hooks fire on every response** — not just task completion. Don't run expensive checks on every response.
4. **Async hooks can't block** — `"async": true` runs in background. Use for fire-and-forget only (logging, linting).
5. **Make scripts executable** — `chmod +x hooks/*.sh`. Use absolute paths: `"$CLAUDE_PROJECT_DIR"/.claude/hooks/...`
6. **statusLine and fileSuggestion** hooks require workspace trust (security fix in v2.1.51).
7. **Debug hooks** — `claude --debug "hooks"` shows hook execution details.

Version History

Version	Date	Change
v2.0.x	Late 2025	Basic hook system introduced
v2.1.0	Jan 2026	Hooks in skill/agent frontmatter
v2.1.47	Feb 2026	Hook execution stability fixes
v2.1.49	Feb 2026	SubagentStop includes <code>last_assistant_message</code>
v2.1.50	Feb 2026	<code>WorktreeCreate</code> , <code>WorktreeRemove</code> , <code>ConfigChange</code> events
v2.1.51	Feb 2026	<code>statusLine/fileSuggestion</code> require workspace trust
v2.1.76	Mar 2026	<code>PostCompact</code> , <code>Elicitation</code> , <code>ElicitationResult</code> events added